Microsoft

GH-300: Labs



[GitHub Copilot Fundamentals Part 1 of 2]

https://www.youtube.com/watch?v=_HBCwxJmq7Y

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Labs (1 to 6)

Exercise 1
Set up GitHub Copilot to work with Visual Studio Code
https://learn.microsoft.com/en-us/training/modules/advanced-github-copilot/3-exercise-setup?ns-enrollment-type=learningpath&ns-enrollment-id=learn.github-copilot
Exercise 2 (Exercise)
Scale institutional knowledge using Copilot Spaces
https://github.com/skills/scale-institutional-knowledge-using-copilot-spaces
Exercise 3
Update a web API with GitHub Copilot (done in Exercise 1)
https://learn.microsoft.com/en-us/training/modules/advanced-github-copilot/5-exercise-update-a-web-api?ns-enrollment-type=learningpath&ns-enrollment-id=learn.github-copilot
Exercise 4
Develop unit tests using GitHub Copilot
https://microsoftlearning.github.io/mslearn-github-copilot-dev/Instructions/Labs/LAB_AK_04_develop_unit_tests_xunit.html
Exercise 5
Expand your team with Copilot coding agent
https://github.com/skills/expand-your-team-with-copilot/
Exercise 6
Integrate MCP with GitHub Copilot
https://github.com/skills/integrate-mcp-with-copilot/

Labs (7 to 12)

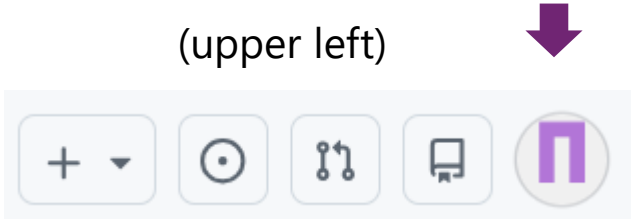
Exercise 7
GitHub Copilot code review
https://github.com/skills/copilot-code-review
Exercise 8
Update a JavaScript portfolio with GitHub Copilot
https://learn.microsoft.com/training/modules/introduction-copilot-javascript/5-exercise-javascript-portfolio
Exercise 9
Update a Python web API with GitHub Copilot
https://learn.microsoft.com/training/modules/introduction-copilot-python/5-exercise-python-web-api
Exercise 10
Customize your GitHub Copilot experience
https://github.com/skills/customize-your-github-copilot-experience
Exercise 11
Create applications with the Copilot CLI
https://github.com/skills/create-applications-with-the-copilot-cli
Exercise 12
Modernize your legacy code with GitHub Copilot
https://github.com/skills/modernize-your-legacy-code-with-github-copilot

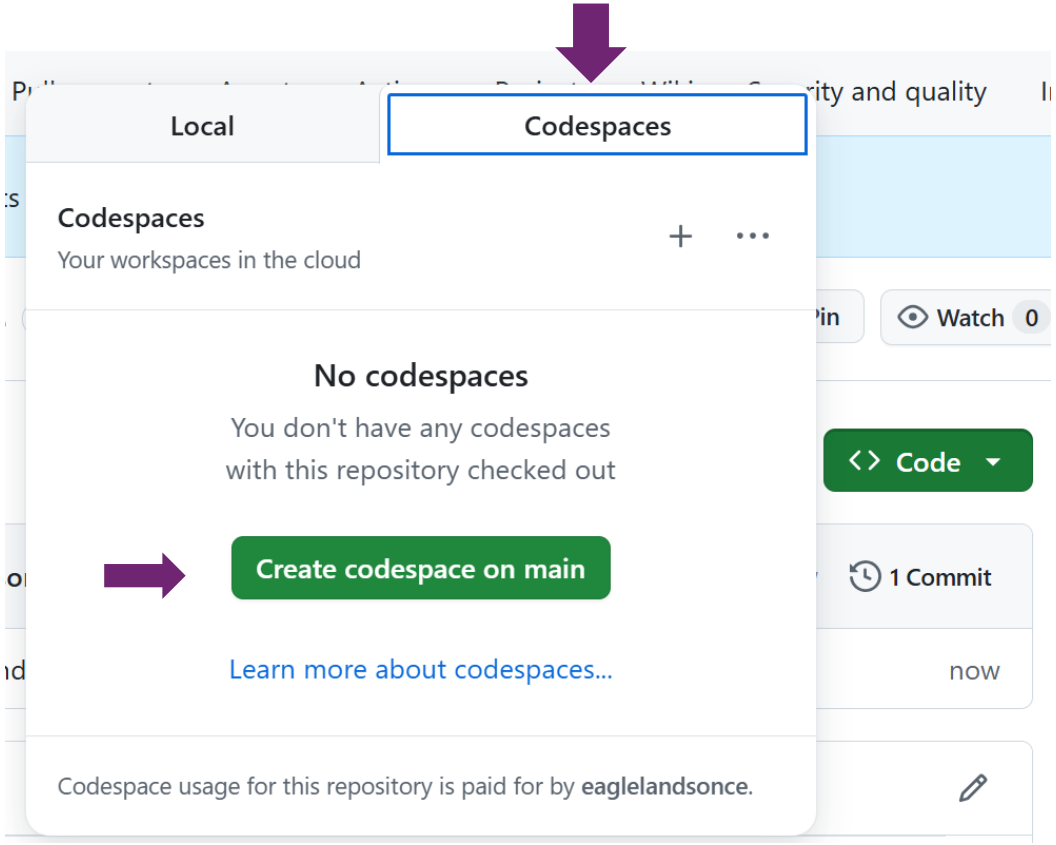
Create your first Codespaces

Up and Running with Codespaces (Get a free GitHub Account!, GitHub)

1. Create your free GitHub Account
2. Create a repository in that account (add a ReadMe file), call it gh300lab
3. Select Codespaces and then Create codespace on main

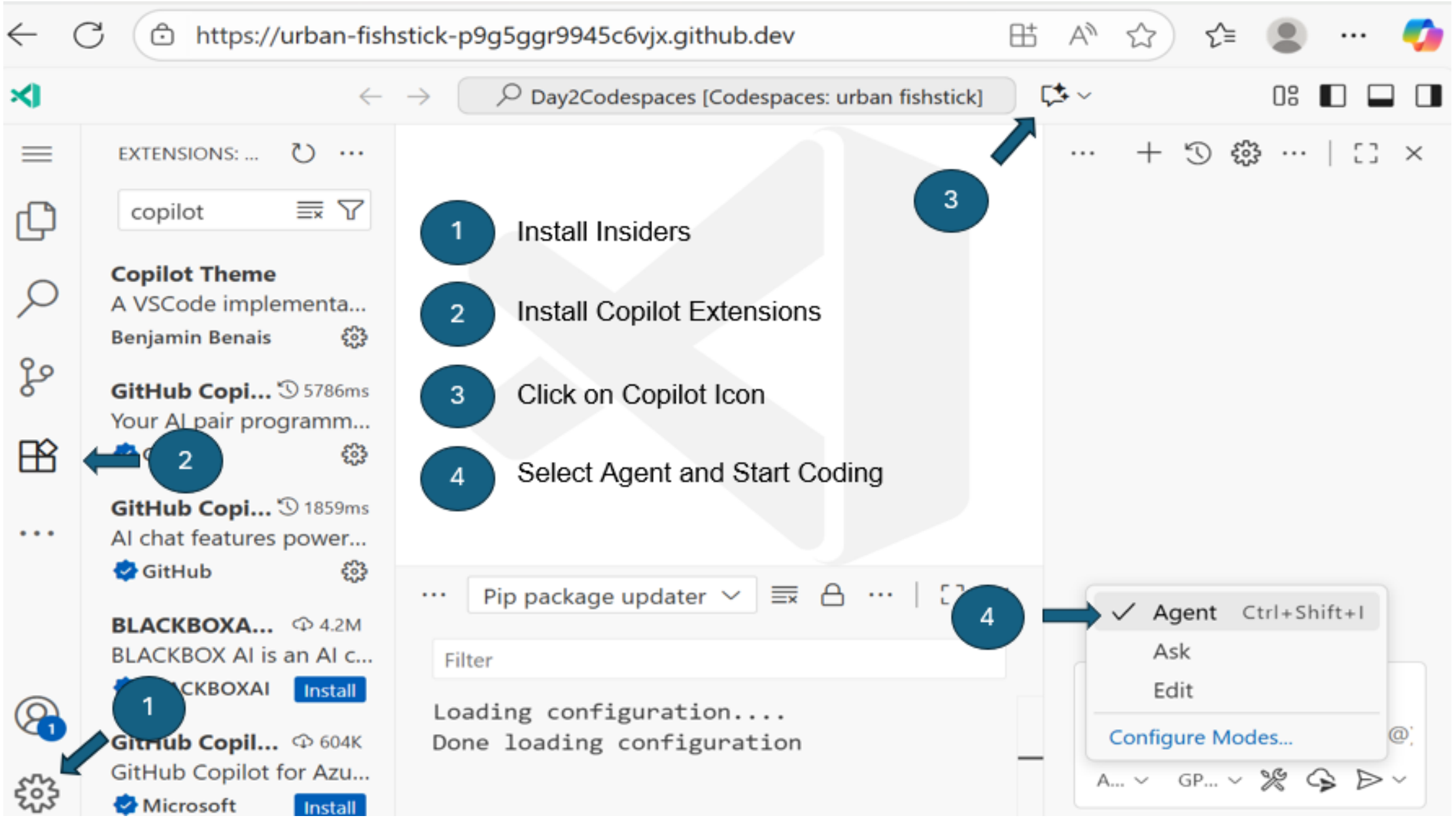
(upper left)





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Codespaces (Same for VSCode)



Getting Started Prompts

GitHub Test Adding Sum of Squares (Codespaces)

1.Create a Python File call it **Lab2Practice.py**

2.Use GitHub Copilot to complete the following functions and test

def sum_squares_of_first_10_integers

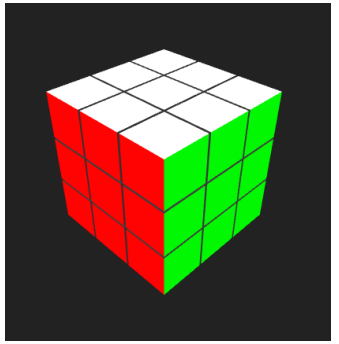
3. In Edit with Copilot (using agent in the chat)

4. Add comments

5. Add unit tests

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Getting Started Prompts



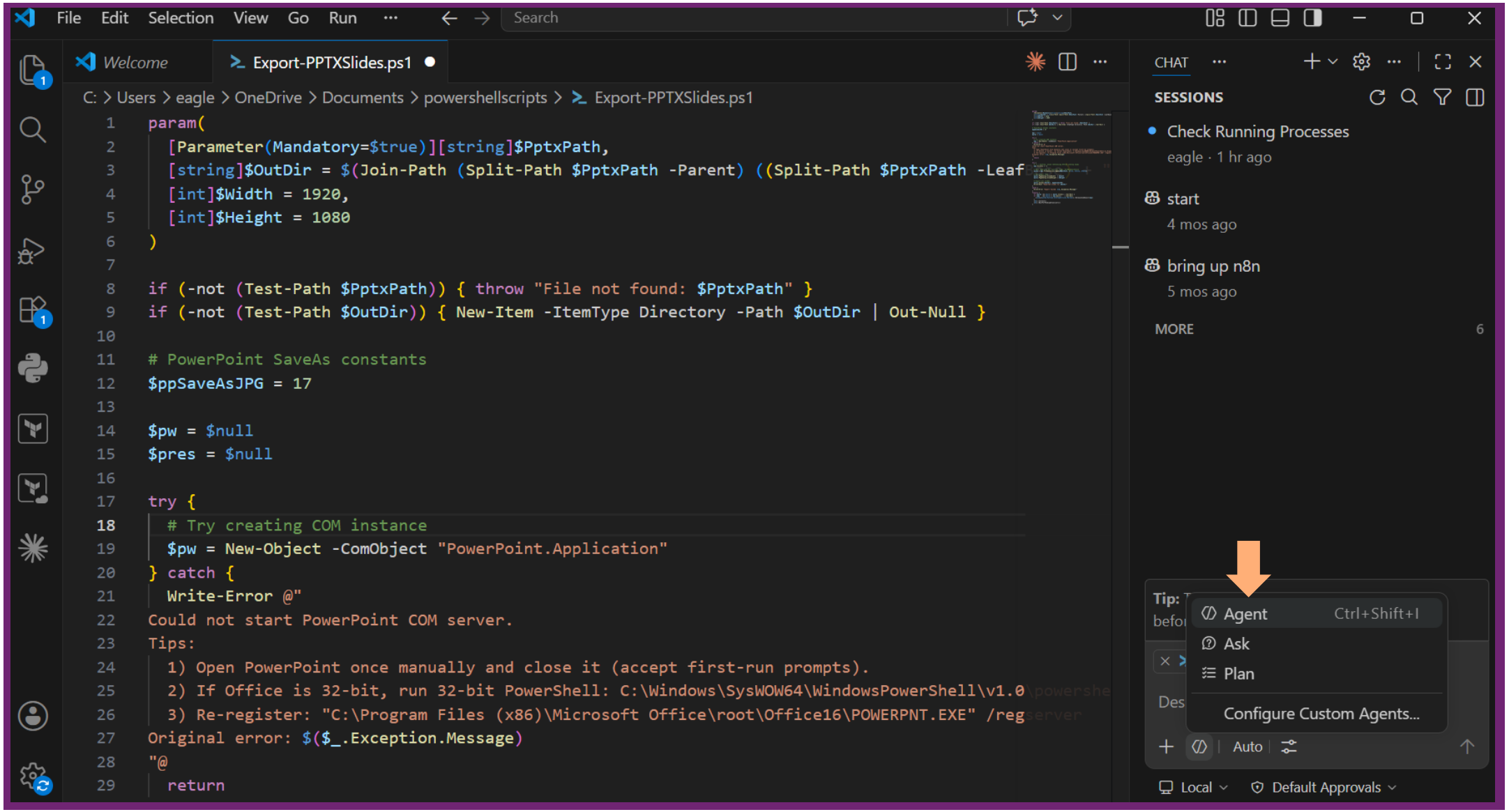
Procedural Prompt: Develop a fully interactive 3D Rubik's Cube simulation using Three.js that visually demonstrates the scrambling and solving of the puzzle in real time. The cube should be constructed using a group of 27 cubelets rendered in a Three.js scene with appropriate coloring and lighting. Users should be able to freely rotate the entire cube with mouse controls, and initiate scrambling using a sequence of randomized legal moves. Implement a visual solving animation using a known algorithm such as Kociemba's or a beginner-friendly layer-by-layer method, where each move is animated with smooth rotations and delays to allow clear step-by-step observation. Include UI controls to scramble, solve, pause, and step through moves. Ensure modular code design to allow for future extensions, such as manual face turns, AI solvers, or integration of group-theoretic state visualization. The final product should serve both as an educational tool and a visually engaging demo of algorithmic problem-solving in 3D --- create index.html, scripts.js and style.css files.

Challenge: when completed add comments and testing.

https://huggingface.co/spaces/eaglelandsonce/HF_Rubric_Cube

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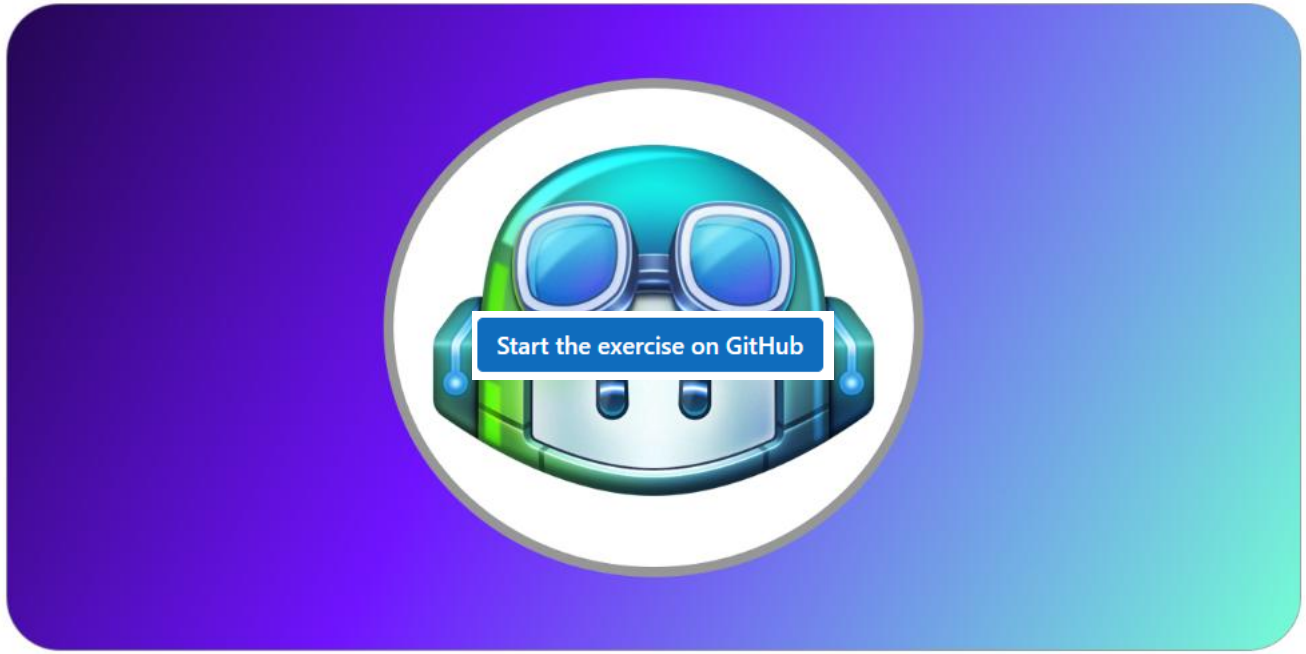
Open in VSCode



https://code.visualstudio.com/download

Exercise - Develop with AI-powered code suggestions by using GitHub Copilot and VS Code

1 minute

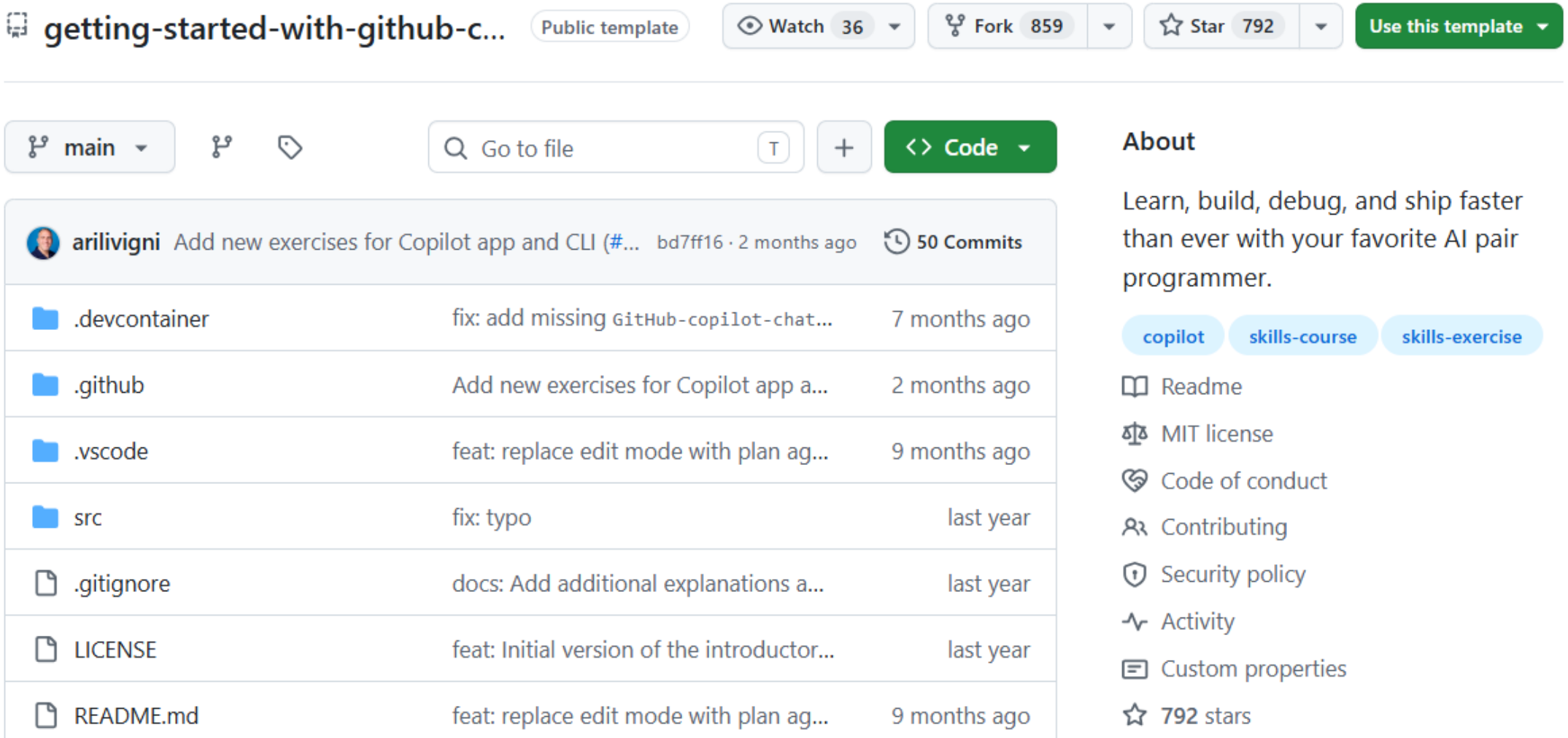


Start the exercise on GitHub

https://learn.microsoft.com/en-us/training/paths/copilot/

https://learn.microsoft.com/en-us/training/modules/introduction-to-github-copilot/

https://learn.microsoft.com/en-us/training/modules/introduction-to-github-copilot/5-exercise



https://github.com/skills/getting-started-with-github-copilot

Demo

GitHub Copilot supports Agent Skills in VS Code agent mode, the Copilot CLI, the Copilot app, the cloud agent, and code review. Copilot loads a skill's instructions and bundled resources when they're relevant to the task. GitHub Docs

To use a skill in a GitHub repository, put its SKILL.md and any supporting files in a folder like .github/skills/songsmith/. The SKILL.md uses YAML frontmatter with a name and description, followed by the workflow instructions. GitHub Docs

Your **SongSmith** is a straightforward candidate because its core workflow is instruction-based.

Your Task: Now create a skill (a joke skill for example) and put it in a file structure like .github/skills/joketeller/ then create a Codespaces and drag it over and using the /joketeller create a joke on any topic you give it.

Course Description

This course introduces GitHub Copilot as an AI-powered coding assistant that helps developers write, understand, test, refactor, and review code more efficiently. Learners explore how to use Copilot responsibly across common development environments, including Visual Studio Code, GitHub.com, chat, and the command line, while also learning prompt engineering techniques, customization options, unit testing support, pull request assistance, agent mode, cloud agent capabilities, and language-specific use cases in JavaScript and Python.

My Goal --- Hands-on Codespaces (GitHub and VSCode) and GitHub CLI

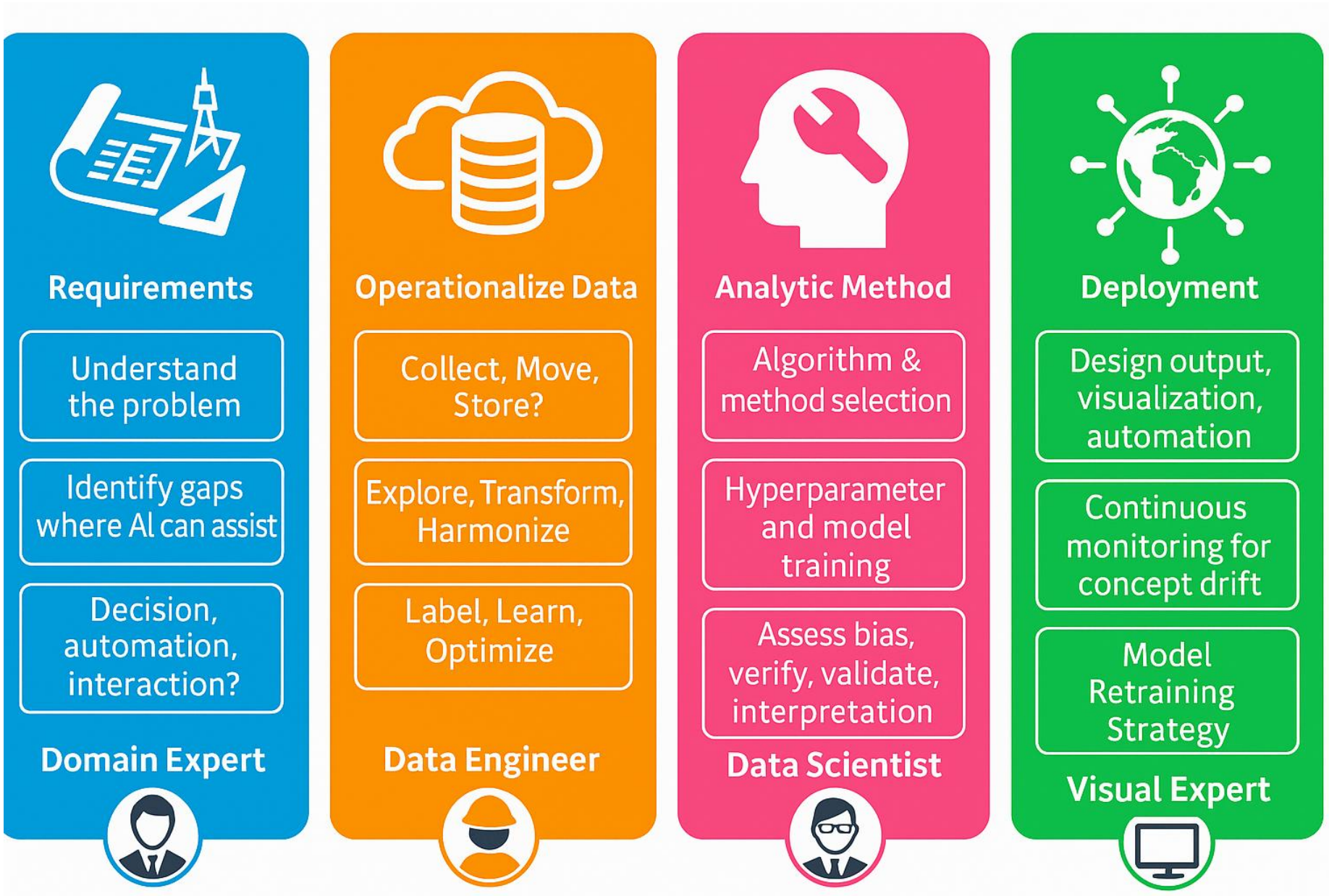
What's Next Thinking





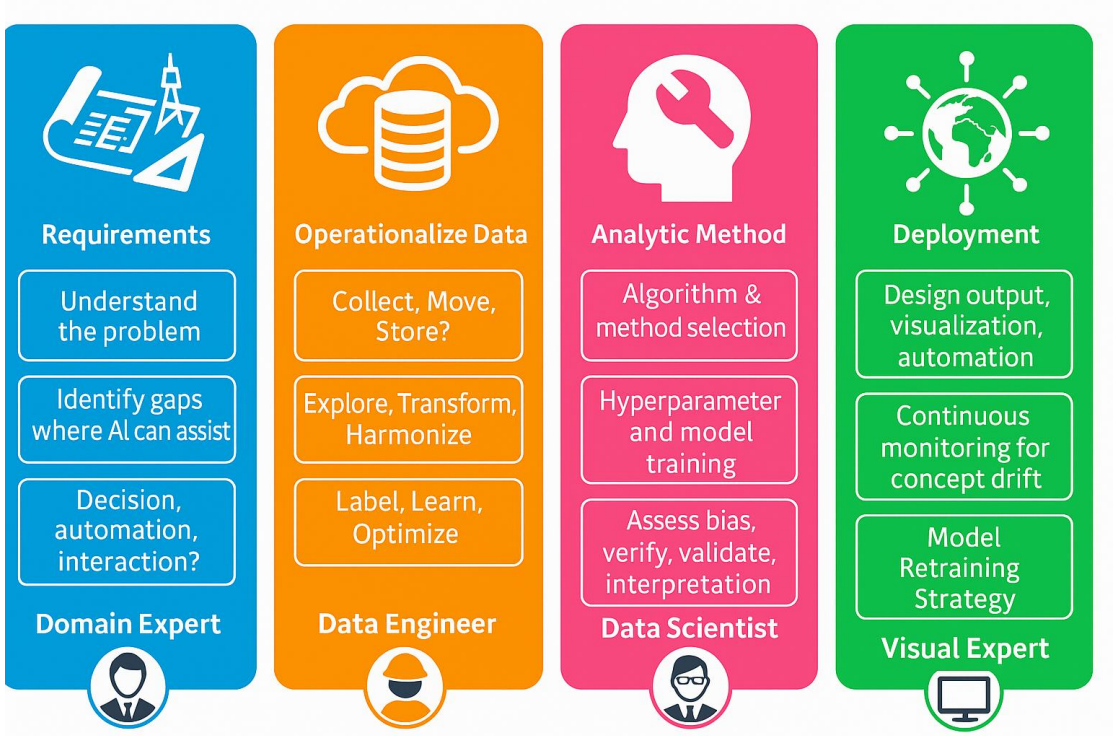
What is the Prompt?

ROAD (Business Plan Creator)



Prompt Exercise

How would you prompt the others?



Requirements

Role: Domain Expert
Name: Dr. Emily Carter
Goal: Identify AI-assisted opportunities by understanding business requirements and decision-making needs.
Backstory: A seasoned business strategist with expertise in AI applications. She has worked with Fortune 500 companies to integrate AI into their workflows, identifying gaps and optimizing automation opportunities.
Task: Analyze Business Requirements for AI Implementation
Description:
•Understand the problem statement and business context.
•Identify gaps where AI can enhance processes.
•Determine the level of automation, decision-making, and interaction required.
•Collaborate with technical teams to align AI applications with business goals.
Expected Output:
A structured report outlining the business requirements, potential AI applications, and recommendations for decision-making strategies.

ROAD Agentic Workforce

Requirements	Operations	Algorithms	Design

Role: Domain Expert
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Expected Output:
A structured report outlining the business requirements, potential AI applications, and recommendations for decision-making strategies.

Role: Data Engineer
Name: Alex Martinez
Goal: Ensure seamless data collection, transformation, and storage for AI model training and analysis.
Backstory: A data pipeline expert with a background in distributed computing and cloud data solutions. Previously worked at a major tech firm designing large-scale data ingestion pipelines.
Task: Data Pipeline Optimization and Engineering
Description:
•Collect and move data from various sources into a unified storage system.
•Explore, transform, and harmonize structured and unstructured data.
•Label, preprocess, and optimize datasets for AI model consumption.
•Ensure data integrity and security within cloud or on-prem environments.
Expected Output:
A documented ETL pipeline, data schema definitions, and a ready-to-use dataset for AI model training.

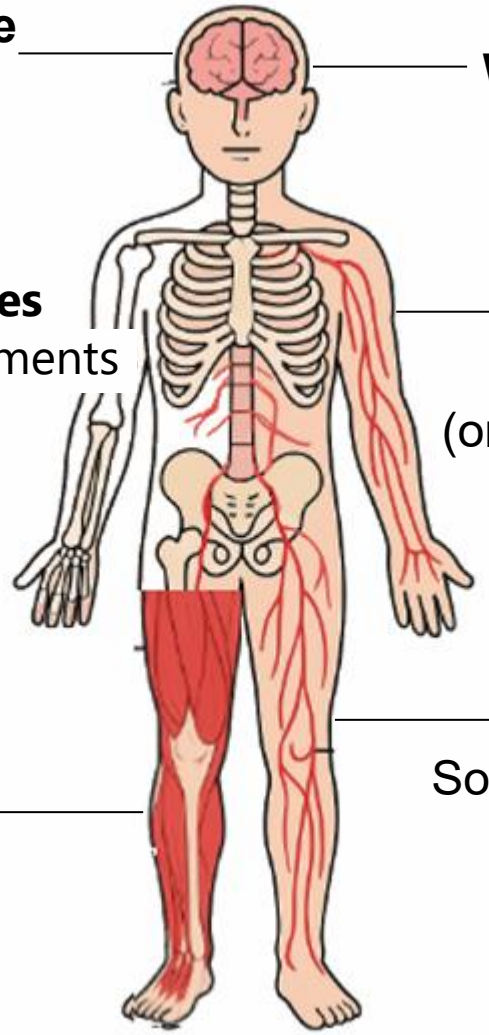
Role: Data Scientist
Name: Dr. Rachel Nguyen
Goal: Develop and validate AI models through algorithm selection, hyperparameter tuning, and bias assessment.
Backstory: A machine learning researcher with a PhD in artificial intelligence. Worked on predictive analytics for healthcare and financial systems, ensuring models are both effective and fair.
Task: Model Development and Performance Optimization
Description:
•Select the best algorithms and methods for the given AI task.
•Tune hyperparameters for optimal model performance.
•Assess bias, verify outputs, validate model generalization, and interpret results.
•Work closely with engineers to integrate the model into production.
Expected Output:
A trained AI model with performance metrics, bias analysis, and validation results.

Role: Visual Expert
Name: Grace Watanabe
Goal: Deploy AI solutions with effective visualization, monitoring, and retraining strategies.
Backstory: A data visualization and UX specialist with experience in AI product design. She has helped companies develop AI dashboards and monitoring tools for real-time decision-making.
Task: AI Deployment and Monitoring
Description:
•Design model output, visualization strategies, and automation workflows.
•Implement continuous monitoring for concept drift and model degradation.
•Develop a model retraining strategy to ensure sustained performance.
•Create user-friendly dashboards to communicate AI insights effectively.
Expected Output:
An AI deployment blueprint with visualization components, monitoring dashboards, and a retraining strategy.

Prompt Engineering

Think of a prompt as the human body

Goal
- Context
  - Requirements
    - Expectations & Functions
      - Data & Sources
        - Examples



Frontal Lobe
Goal

Working Memory
Context

Bones
Requirements

Skin
Examples
(one shot, two shot)

Blood
Sources and Data

Muscles
Expectations and Functions

Use Case



A local hardware store faces significant challenges in managing the billing process for customers. Currently, the store uses a manual checkout system, which is time-consuming, prone to human errors, and inefficient. This often results in customer dissatisfaction and potential revenue loss due to mismanagement or inaccurate billing.

To address this, the hardware store is exploring the implementation of a basic Cart Management System as a pilot solution to improve the purchasing experience. The system does not need to be overly complex but should provide essential functionalities to streamline operations and enhance customer satisfaction.

https://huggingface.co/spaces/eaglelandsonce/Hardware_Store

Example of a multi-modal prompt
(Create in Codespaces)



Goal and Context: The primary goal is to create a program that digitally streamlines the billing process for a local hardware store that currently relies on manual methods, increasing efficiency, enhancing customer satisfaction, reducing operational errors, and laying the foundation for future growth.

Data

```
Inventory = [
    (101, "Hammer", 120),
    (102, "Screwdriver Set", 350),
    (103, "Drill Machine", 2500),
    (104, "Pliers", 180),
    (105, "Wrench", 220),
]

My_Cart = []
```

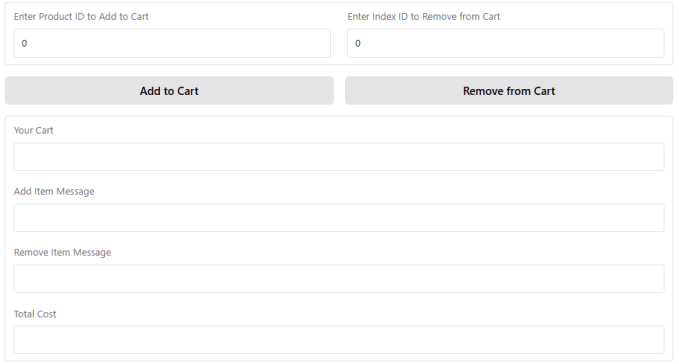
Requirements:

Add Items to the Cart: The store clerk should be able to select an item from the inventory and add it to the cart. The item's name and price should be pulled automatically based on the selected ID.
Remove Items from the Cart: The clerk should have the flexibility to remove individual tools from the cart before checkout to accommodate customer preferences or mistakes.
Calculate Total Bill: The system should calculate the total cost of items in the cart and display the final amount clearly to the customer for quick payment.

Expectations and Functions

The Gradio interface would include: A dropdown to select tools from inventory
An "Add Tool" buttonA second dropdown to select tools from the current cart
A "Remove Tool" buttonA display box showing the cart contentsA "Calculate Total" button with a result output

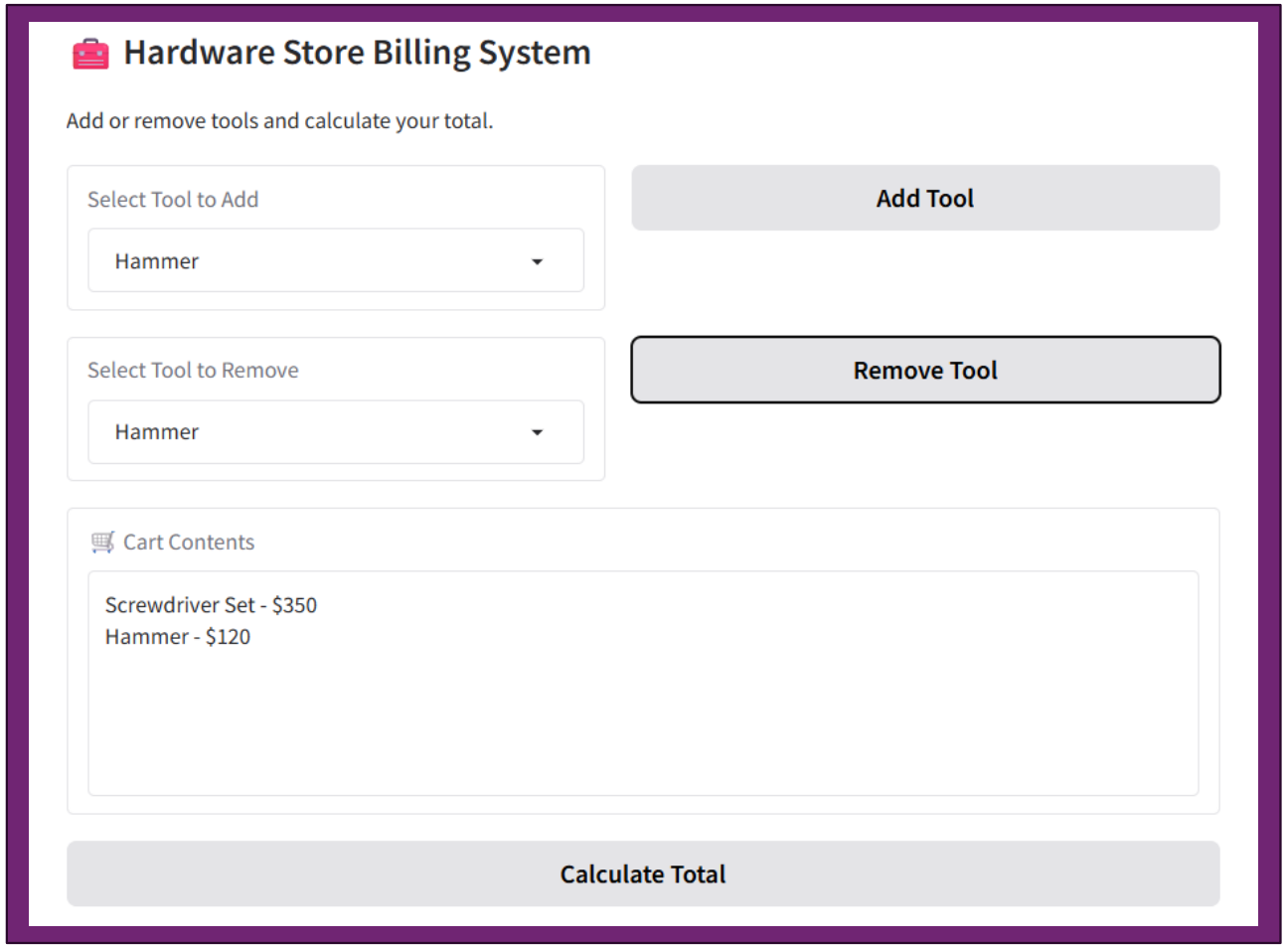
Interface Example



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Use Case

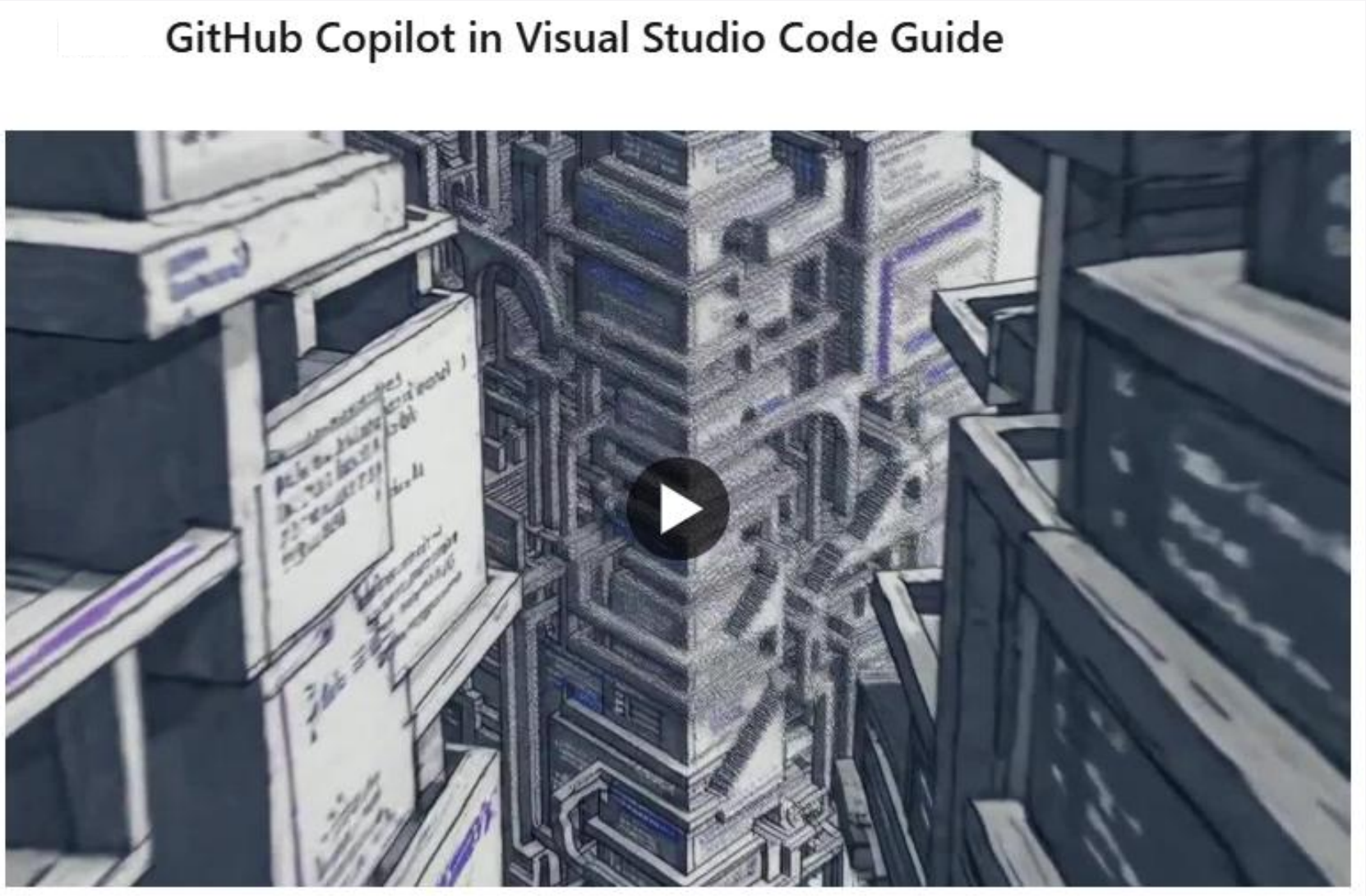




https://huggingface.co/spaces/eaglelandsonce/Hardware_Store

https://www.linkedin.com/pulse/gh-300-under-construction-michael-lively-k0gqe/

Lab



https://www.linkedin.com/pulse/gh-300-under-construction-michael-lively-k0gqe/

Set up GitHub Copilot to work with Visual Studio Code



GitHub Copilot can help you code by offering autocomplete-style suggestions right in VS Code and Codespaces.

Get a personal GitHub account (www.GitHub.com)

https://learn.microsoft.com/en-us/training/modules/advanced-github-copilot/3-exercise-setup?ns-enrollment-type=learningpath&ns-enrollment-id=learn.github-copilot



https://www.youtube.com/watch?v=VkOibxsQ1oU

Exercise 1 – Install and test Copilot CLI



Install copilot CLI, authenticate, install PowerShell, examine running services.

In this exercise, you do the following:

1. Check to see if you have node installed (if not install it) https://nodejs.org/en/download
2. node –version
3. **Install copilot cli:** npm install -g copilot
4. **Run copilot:** copilot
5. **Ask a question:** what process do I have running on my machine
6. If necessary login to copilot /login (and authenticate)
7. What processes do I have running on my machine
8. Manually Install PowerShell 6+ https://learn.microsoft.com/en-us/powershell/scripting/install/install-powershell-on-windows?view=powershell-7.5
9. (restarts copilot cli) Now you have PowerShell superpowers!!

Exercise 2 – Install and test Copilot CLI





Using Copilot CLI confirm your

In this exercise, you do the following:

1. What processes are running on my machine
2. List the top five CPU users
3. Create a folder in documents name HelloWorldCopilotCLI and a text file in that document named HelloWorldCopilotText and in the text put Hello world from copilot, you now have the power!
4. What is the path to this folder
5. Navigate to the folder and confirm the results
6. Can you turn this into a git repository???

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Exercise – Build a Jeopardy Compete Game



Build a Jeopardy Game where two teams (A and B) go against each other and the team with the highest score wins. Add comments and unit test, use Jest as the JavaScript testing package.

- Build a Jeopardy Game where two teams (A and B) go against each other and the team with the highest score wins. Add comments and unit test, use Jest as the JavaScript testing package.

- 1. Create a repository (JeopardyCompete) and create a Codespaces from that repository
- 2. Use the reference below as starter code.
  Reference:
  https://huggingface.co/spaces/eaglelandsonce/Jeopardy_Six_Prompting
- 2. Craft a prompt for your Agent and vibe code the project.
- 3 After the game is completed and running successfully add comments and JavaScript test - using he Jest package.



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Exercise - Set up GitHub Copilot to work with Visual Studio Code



In this exercise, we create a new repository using the GitHub template for the JavaScript personal portfolio frontend web application.

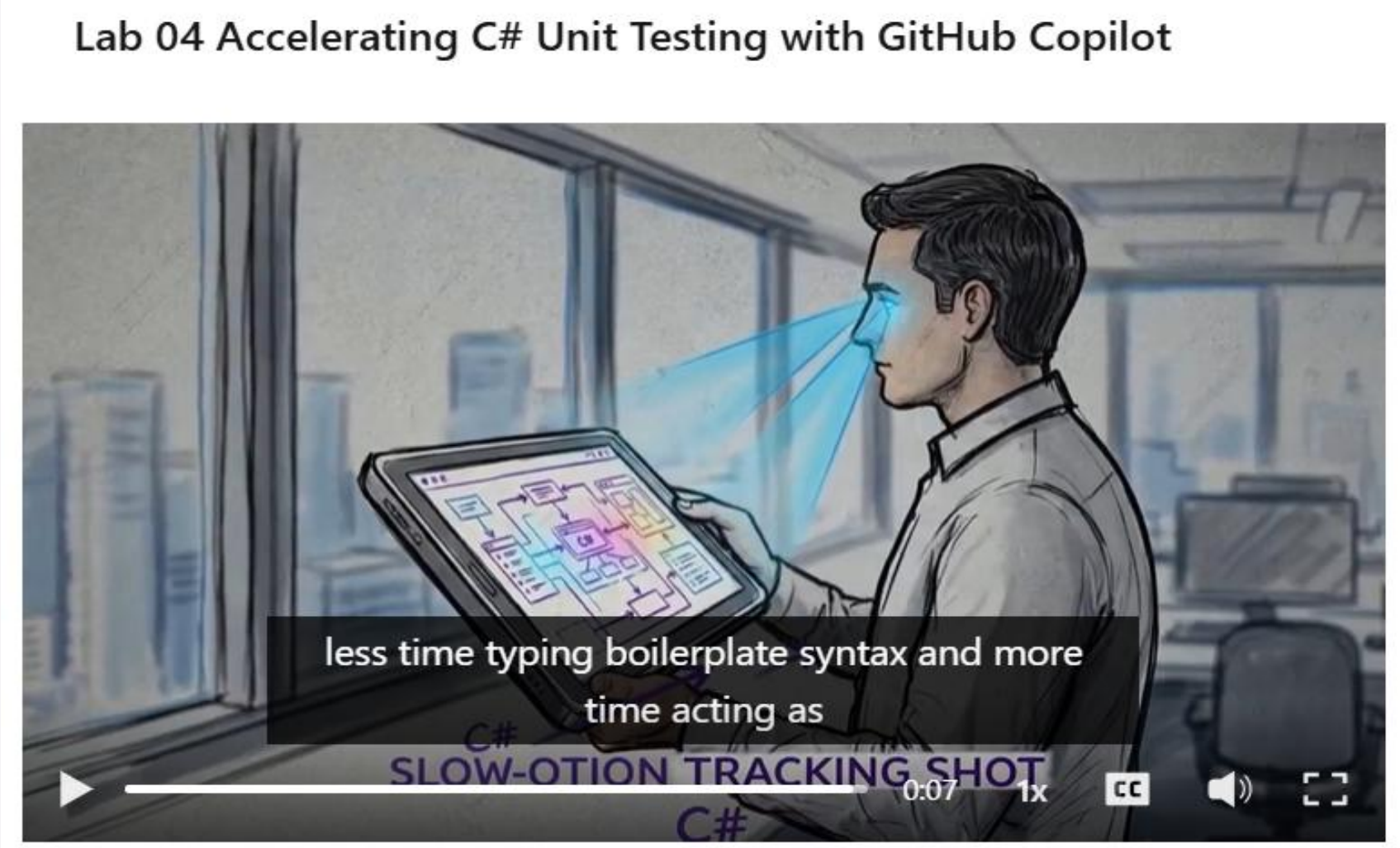
In this exercise, you do the following:

- **GitHub Account**: Create a GitHub account if necessary, as it's required to use GitHub Copilot.
- **Check Copilot Access**: Review pricing plans and ensure you have access to the appropriate plan, such as Copilot Free for learning.
- **Verify Access**: In GitHub settings, ensure Copilot and Copilot Chat are enabled.
- **Install Extension**: Install the GitHub Copilot extension in your IDE from the extension marketplace.

Exercise - Set up GitHub Copilot to work with Visual Studio Code

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Lab



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https://learn.microsoft.com/en-us/training/modules/develop-unit-tests-using-github-copilot-tools/6-exercise-develop-unit-tests-github-copilot

Exercise - Update a JavaScript portfolio with GitHub Copilot



In this exercise, you add animations with live suggestions and use a prompt to customize scroll behavior from an already existing JavaScript template repository.

In this exercise, you do the following:

- **Portfolio Customization**: Use a React-based web application template to customize your portfolio with personal information in the siteProps variable in src/App.jsx.
- Add an animation to your social media icons by prompting GitHub Copilot in src/styles.css to generate CSS code for a hover effect.
- Accept Copilot's CSS suggestions by pressing the tab key, or refine the code until it matches your desired animation.
- Your site will automatically reload in the Codespace, allowing you to see the animation effect by hovering over the social media icons in the footer.

Exercise - Update a JavaScript portfolio with GitHub Copilot

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